PROJECT DOCUMENTATION

- **Team Name : CodeSib**
- **Team Leader : Yukta Chaudhari**
- **Problem Statement chosen** : PS03
- **Event name:** HackOcean 2026 - Round 2
- **Links**
  GitHub Repository link : Aditi-179/OceanSearch
  Live Deployed Link : https://oceansearch-opal.vercel.app/

- **Project Name/Title :** OceanSearch
- **Summary:** OceanSearch is a highly interactive, scrollytelling web application that takes users on a visual descent through the ocean's depth zones. It blends a breathtaking 3D WebGL underwater ecosystem with real-time UI overlays that simulate AI-driven marine tracking, environmental IoT telemetry, and marine life discovery to create a next-generation educational platform.
- **Problem being solved :** Traditional environmental data (such as coral bleaching, microplastic accumulation, and overfishing) is often presented in static, dry reports that fail to emotionally engage the general public. This project bridges that gap by visualizing ocean health crises and marine biology in a highly immersive, game-like digital experience, fostering empathy, deep education, and conservation action.
- **USP (Unique Selling Point)**
- **Key Features :**

- **Tech Stack :**

Core Framework: Next.js (App Router), React 19
3D / WebGL: React Three Fiber, @react-three/drei, Three.js
Styling: Tailwind CSS v4
Animations & Scrolling: GSAP, Lenis (for smooth scrolling), Framer Motion
Data Visualization & Mapping: Recharts, Mapbox GL, React-Map-GL
Icons: Lucide React, Phosphor Icons
Real-time Comms: Socket.io-client

- **Future Scope :**
  1. Backend AI integration
  2. Live underwater drone feeds
  3. Real-time IoT sensor connectivity
  4. AI-powered marine species detection
  5. Donation payment gateway
  6. Multi-language support
  7. Progressive Web App (PWA)